EXISTENCE OF K-FLAT RESOLUTIONS WITH FLAT TERMS AND TERMWISE SURJECTIVE MAPS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction and verification, both inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complex and the construction of resolutions by generators. Same chapter, the characterization of K-flatness and the construction of complexes mapping onto given complexes with controlled terms. Supplies the construction of the resolution in the proof.
- Flat modules and the existence of surjections from flat modules. Same chapter, the existence of surjections from flat modules onto arbitrary modules, which makes the termwise construction possible. Supplies the termwise surjectivity and flatness of the terms.

Source reference. Original lemma 077J: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3173–3173. Navigation pointer invoked at proof line 3216; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03EV: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3219–3219. Navigation pointer invoked at proof line 3219; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 06YQ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3120–3120. Navigation pointer invoked at proof line 3221; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YR: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3152–3152. Navigation pointer invoked at proof line 3223; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. For any complex $\mathcal{G}^\bullet$ there exists a K-flat complex $\mathcal{K}^\bullet$ whose terms are flat $\mathcal{O}$-modules and a quasi-isomorphism $\mathcal{K}^\bullet \to \mathcal{G}^\bullet$ which is termwise surjective.*

Proof. Choose a diagram as in Lemma 077J. Each complex $\mathcal{K}_n^\bullet$ is a bounded above complex of flat modules, see Modules on Sites, Lemma 03EV. Hence $\mathcal{K}_n^\bullet$ is K-flat by Lemma 06YQ. Thus $\operatorname{colim} \mathcal{K}_n^\bullet$ is K-flat by Lemma 06YR. The induced map $\operatorname{colim} \mathcal{K}_n^\bullet \to \mathcal{G}^\bullet$ is a quasi-isomorphism and termwise surjective by construction. Property (3) of Lemma 077J shows that $\operatorname{colim} \mathcal{K}_n^m$ is a direct sum of flat modules and hence flat which proves the final assertion. □